



Department of Electrical and Computer
Engineering
Instructor : Hassan Tfaily

Lab 7 : “LCD”

Introduction and objectives

Liquid Crystal Displays (LCDs) are widely used in embedded systems to visually convey information such as sensor readings, text messages and data outputs. In this lab, you will gain hands-on experience in sending data to the LCD, implementing basic display functionalities and programming a scrolling text.

By the end of this lab, you should be able to:

Explain how an LCD operates and interfaces with a microcontroller.

Write and test programs that display characters, strings and numbers on the LCD.

Troubleshoot common software issues in LCD interfacing.

Write fully commented programs for the PBMCUSLK board, including appropriate labels, memory operands and constants, to solve the 2 tasks below:

Task 1:

The first program should do the following:

- print an integer number, two- digits, on the LCD.
 - Initialize the LCD as we did in Lab6.
 - The number is created as a variable using define constant byte **dc.b**.
 - To get the ASCII equivalent of a decimal number, add this number with the value #48.
 - Use the instruction ‘16 by 16 integer divide’ in the **instruction_set** datasheet so that you can print the TENS digit , then the ONES digit on the LCD.

- Using the previous code, display a digital clock on the LCD as shown below:

minutes : Seconds
00:00 → 00:59
01:00 → 03:59

The 'seconds' count from 0 till 59 → the 'minutes' increment by 1 → the 'seconds' will be reset to 0 to restart counting again.

Task2:

The second program should do the following:

Print the word 'Embedded' on the LCD by following these steps :

- initialize the LCD
- declare a table of char to define the word 'Embedded' and use the index register in order to print it.

Task3:

The third program should do the following:

- print the word 'COE' in the middle of the first line.
- print the scrolling message : 'Embedded System' on the second line of the LCD.

Make sure to follow these steps:

- Define the table of characters and all the needed variables.
- print the characters 8 by 8 since the LCD display is 2*8.
- After printing the first 8 characters, restart with the next character in the message and print the successive 8 chars.
- During the print of each successive 8 characters, if you have reached the last character of the message, restart from the beginning.
